

Dilara Caglar

Project Portfolio 2026

Carnegie Mellon University

Anticipated Graduation - May 2028

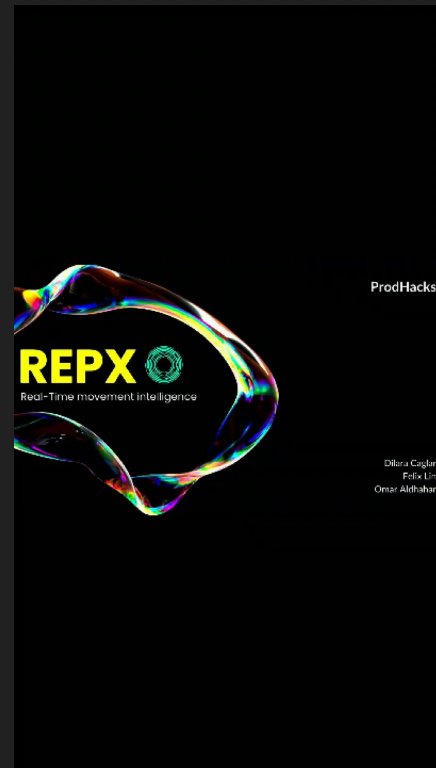
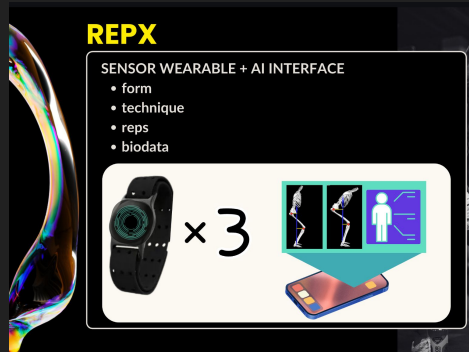
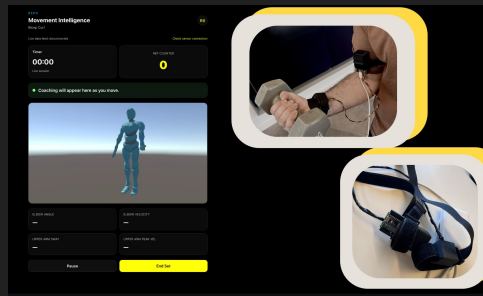
B.S. Electrical and Computer Engineering (ECE)

Additional Major: Engineering Design, Innovation & Entrepreneurship (EDIE)

REPX — Real-Time Movement Intelligence

A hardware-software ecosystem providing objective biomechanical feedback for strength training.

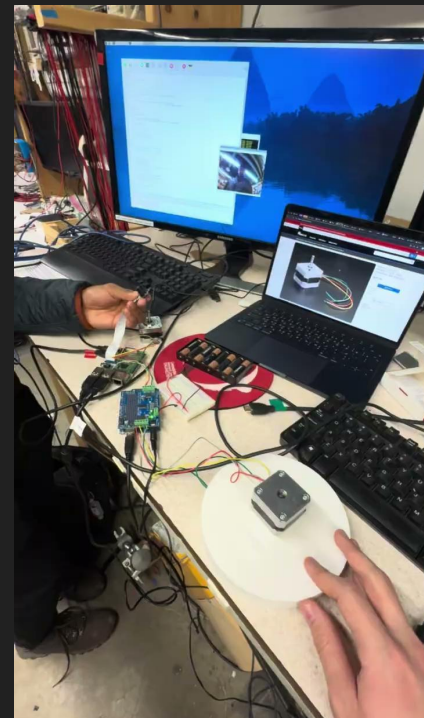
- **Wearable Sensor Array:** Developed a multi-node IMU system (arm/wrist) to capture **high-fidelity** joint kinematics.
- **Real-Time Data Pipeline:** Streamed spatial data via Bluetooth/Serial to a custom **React** dashboard.
- **Biomechanical Metrics:** Calculated real-time elbow angle, velocity, and "Upper Arm Sway" to identify form breakdowns and momentum cheating.
- **Live Digital Twin:** Implemented a 3D skeletal rig that mirrors user movement for immediate visual correction.



Build18 — Autonomous Tracking Fan

An intelligent fan using computer vision to track and follow a user in a workspace.

- **Embedded CV:** Integrated a **PyTorch-based** object detection model to identify and localize the user within the frame.
- **Closed-Loop Control:** Engineered a **PID controller** on Arduino to translate visual coordinates into smooth stepper motor actuation.
- **Rapid Prototyping:** Designed and 3D-printed a custom dual-axis gimbal and gear assembly for the fan chassis.
- **Performance:** Achieved low-latency tracking, ensuring the airflow remains centered on the target during lateral movement.



Slide 3: IoT Transit Matrix & Kinetic Iris

- **Integrated Hardware Stack:** Engineered a high-density display system using the **Adafruit Matrix Portal S3** paired with dual **64x32 RGB LED panels**.
- **System Scalability:** Engineered a daisy-chain configuration using IDC ribbon cables to expand the resolution to a unified display.
- **Firmware Environment:** Developed using CircuitPython to leverage the Adafruit library, enabling advanced features like multi-panel chaining
- **Power Management:** Designed a high-current power distribution system utilizing a 5V 10A switching power supply and screw terminal adapters to drive 4,096 pixels concurrently.
- **Software Logic:** Programmed custom scrolling text, fonts, and color-coded destination alerts with an imitation timer for each location.



Logic Gates & Integrated Circuit Shield

Custom PCB Design & Hardware Validation

- **PCB Development:** Designed and assembled a custom **CMU ECE Arduino Shield** specifically for the **18-021** (Spring 2025) systems course.
- **Logic Integration:** Engineered on-board hardware to demonstrate fundamental Boolean operations including **AND**, **OR**, **NAND**, and **NOR** gate logic.
- **Component Assembly:** Executed precision Surface Mount Device (**SMD**) soldering for integrated circuits (**U1, U2**), transistors (**QA-QB**), and various passive resistor networks (**R1-R10**).
- **Power & Signal Routing:** Implemented level-shifting and voltage regulation traces to ensure signal integrity between the **Arduino Uno R3** base and specialized logic components.
- **Hardware Validation:** Performed end-to-end circuit testing to verify truth table accuracy and timing characteristics for complex logic expressions.

